ABSOBEHA ADAMU

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PROFESSIONAL SUMMARY

Operated as an Associate Software Engineer for 2 years and completed my Bachelor's Degree in Electrical Engineering and Minor in Computer Science from Carleton University. I have also engaged in a variety of other roles, projects and experiences that required me to be a driven, punctual and cooperative teammate.

EDUCATION

BACHELOR OF ELECTRICAL ENGINEERING, MINOR IN COMPUTER SCIENCE

Carleton University / Graduation February 2022

- Awarded Engineering Entry Scholarship

OBJECTIVE

To secure a position at your company, to expand my skills, experiences and qualifications.

COMPLETED COURSES & CERTIFICATIONS

Ultimate Shopify Tutorial (2025)

Issued By: UDEMY

Completion Date: Aug 2nd, 2025

Graduated from one of the most thorough Shopify Training programs designed to create and grow eCommerce sites that resulted in conversions. Developed real-world skills in store building, branding, customer-focused design with GemPages (no coding), and writing persuasive copy with human qualities via AI tools like ChatGPT Created my first ever full fledged Shopify store, then set up a fully functional Shopify store where I integrated email marketing thru Omnisend and did some basic SEO on the website. The perfect training to help you get a product based online business up and running —or help run an existing one even better with a focus on design, functionality or conversion.

Certificate ID: UC-6f1f35ab-de61-43ae-8e9f-1abf803349a5

Power BI 101 – Microsoft Power BI Essentials & Certification

Issued By: UDEMY

Completion Date: July 26th, 2025

Certified in Microsoft Power BI through a business intelligence, data visualization and analytics project based course. Mastered data modeling, transformation and visualization by connecting multiple data sources to design dashboards that were interactive, dynamic while they aid in decision making. Communicated data-driven insights effectively using best practices from industry And this training laid a

strong Power BI understanding, enabled analytics in real life scenarios for Business, Financial and Operations.

Certificate ID: UC-562d9157-2c80-49d8-bc7a-73381b973489

Docker Deep Dive: Build, Ship, and Run Containers

Issued By: UDEMY

Completion Date: July 25th, 2025

Gained experience with tools and practices for working within a modern DevOps workflow with Docker, including building images and containers, networking, container storage (volumes), orchestration (docker-compose), JavaScript Task Runners such as Grunt/Gulp. Managed to build custom Docker images using Dockerfiles, run multi-container apps with Docker Compose and work with orchestration tools like Docker Swarm and Kubernetes. Gained hands-on experience setting up container networks, ensuring deployables are secure and automated environments to help in scale applications. Useful for building scalable, modern applications and cloud-native infrastructures where continuous deployment of updates is preferable;element developers and IT professionals that are interested in streamlining the creation of development pipelines and optimizing the infrastructure forState in cloud native environments.

Certificate ID: UC-6cac1ad7-bbbe-4899-9607-446644a7dbe5

Mastering Linux: The Complete Guide to Becoming a Linux Pro

Issued By: UDEMY

Completion Date: June 17, 2025

Took a full-spectrum Linux training, graduated as an excellent system administration in command-line proficiency, shell scripting, permissions, networking and performance monitoring. Developed strong hands-on experience managing users, automating administrative tasks, configuring firewalls, and troubleshooting systems across a wide range of Linux distributions Can comfortably use Linux Operating Systems — getting you ready for System Administration, DevOps, and Infrastructure Engineering roles. Good understanding of open-source technologies and basic networking concepts.

Certificate ID: UC-d9d04da7-080c-40d4-bd23-ee136abd007a

Mastering PostgreSQL: The Ultimate SQL Guide for Beginners

Issued By: UDEMY

Completion Date: June 8th, 2025

Description: Completed the *Mastering PostgreSQL: The Ultimate SQL Guide for Beginners* course on UDEMY. The course provided me with an immersive, engaging and descriptive guide to mastering PostgreSQL and pgAdmin. The course covered the core concepts such as data manipulation, data retrieval, managing advanced clauses, and manipulating tables in various ways. The course included topics on mathematical and logical operations, case statements, data transformation, using functions and maintaining data integrity. The program also recognized the importance of schema design, renaming and altering tables and identifying hidden data relationships through joins and query analysis, acknowledging the importance of these skills when referring to data analysis. Developed practical skills for analyzing and managing datasets effectively.

Certificate ID: UC-496c6e33-3426-46e3-82f8-22d15a595603

Cyber Security 101

Issued By: TryHackMe

Completion Date: June 7th, 2025

Description: This program gave me a comprehensive foundation in key areas of cyber security, by means of theoretical knowledge and hands-on experience. Provided a core understanding of networking principles, encryption techniques and protocols for secure data communication. Refreshed on practical skills on Windows and Linux command-line interface. Proficient in using and understanding various penetration testing tools and techniques to recognize vulnerabilities in applications and systems. Obtained knowledge of strategies and tools for protecting systems against attacks, with an emphasis on incident response and defense measures. Engaged in simulated hands-on SOC operations and gained knowledge on their roles and responsibilities for the security of any given network. Also refreshed and refined my SQL fundamentals and web application security, provided insight on SQL syntax, properties and common security risks associated with SQL injections.

Certificate ID: THM-QSCJIPUZM4

Pre Security Path

Issued By: TryHackMe

Completion Date: May 31st, 2025

Description: Immersive, hands-on cybersecurity training certification focused on foundational security concepts and principles. Gained practical experience using their built-in testing platform, engaging with Windows and Linux Operating systems, learning networking concepts and crucial data transmission information. Explored threat identification, ethical hacking basics and attack vectors. Gained skills in Powershell and Linux bash command-line, navigating file systems and how to optimize the use of terminal-based tools. This certificate provided a fantastic foundation for further improvement.

Certificate ID: THM-6HDTAF9TC4

Computer Forensics and Digital Forensics Masterclass PRO +

Issued By: UDEMY - OCSALY ACADEMY

Completion Date: May 30th, 2025

Description: This complete and extensive course covers all fundamental areas of digital, computer and mobile forensics, includes techniques, tools, legal procedures and structures and real-world applications in cybercrime investigations. This course is intended for students, IT professionals and others who may require this information. It equips its students with the skills and knowledge to detect for digital inconsistencies by analyzing digital evidence, demonstrating various methods in doing so.

Certificate ID: UC-cf3334a5-b581-47c7-b8e7-cb4fbf58f8af

WORK EXPERIENCE

ASSOCIATE SOFTWARE ENGINEER (September 2022 - July 2024)

Toronto Dominion (TD) Bank

Front-End Development

- Developed quality and dynamic user interfaces (UI) using JavaScript and React, often for webpages, ensuring a smooth and engaging environment for any user

Back-End Development

- Built and maintained quality scalable backend programs using Java and Node.js, applying API integration as well, so data can be communicated effectively between services

SQL Database Development

- Designed and improved on SQL Databases, implementing APIs to ensure data can be properly communicated

Application Security

- Resolved vulnerabilities using Veracode, correcting a variety of vulnerabilities which are graded based on their level of severity, while adhering to coding standards and improving reliability and security

Linux Proficiency

- Wrote and executed applications in Linux environments, ensuring quality programming and performance

Volunteer Engagement

- Willingly and actively participated in preparing company events, ensuring a high quality environment for guests and doing my best to contribute to a positive workplace culture

Problem Solving Skills

- Demonstrated the ability to solve problems on my own, without any extensive supervision, while also demonstrating the ability to communicate any problem clearly to a superior if resources are being exhausted
- While also assuring proper communication between teammates, providing support where necessary and demonstrating the willingness to provide extra support for those in need

SERVER (August 2021 - February 2022)

East India Company

- Provided excellent service to the customers attending the restaurant
- Maintained a clean and sanitized environment for the customers and fellow workers
- Functioned well in a fast-paced environment, while coordinating with coworkers

TIER 1 FINANCIAL SUPPORT SPECIALIST (June 2021 – August 2021)

Millennium1Solutions

- Was assigned and trained to answer calls in an appropriate and friendly manner
- Proficient using the Salesforce cloud software for data entry purposes
- Learn a variety of softwares in a short period of time, and become competent in a fast-paced environment
- Providing customers with consistent/friendly credit card support, dealing with a variety of issues

SOFTWARE TECHNICIAN SUPPORT (May 2019 – July 2019)

The Ottawa Hospital IT Medix EPIC

- Support and assist at the operational system level of a software that required deep testing, submitted tickets required for verification using ServiceNow
- Reported performance defects using ServiceNow and verified with development teams to verify the bug repairs, and to receive an update on the status
- Developed proficiency & obtained experience using the EPIC database by assisting nurses

and physicians in documenting and organizing patient information, promptly

- Improved hospital work environment by identifying and resolving database technical issues to support the nurses and physicians
- Explain technical/software issues clearly and accurately in a fast-paced environment

YOUTH WORKER (December 2018 – January 2021)

Boys And Girls Club Of Ottawa

- Design educational and engaging programs for the youth
- Responsible for their safety & interactions with the other youth
- Creating an inviting and inspirational environment & keeping a strict attendance record

CAMP COUNSELLOR (July 2015 – August 2017)

Plant Pool Recreation Association (PPRA)

- Provided youth with a safe environment
- Kept documents on the youth's registration and progress over the course of the summer
- Organize and engaged in games, sports, etc. for the youth to enjoy

KITCHEN WORKER (June 2014 – September 2014)

Back Lane Café

- Maintained a clean environment throughout the entire restaurant
- Washed dishes & prepared ingredients for the head chef
- Lifted heavy crates of wood to fuel the fire of the oven

RELEVANT SKILLS, EXPERIENCES, & ACCOMPLISHMENTS

Analytical Skills

- Wrote reports, analyzed data samples, and conducted presentations using Microsoft Word, Excel, and PowerPoint respectively
- Collected data on Analog AM modulation and demodulation, then generated plots using Excel and MATLAB & Simulink for a final report & read schematics and designed using CAD tools
- Developed the required skills in economic analysis to make sound economic and financial decisions to improve the efficient use of available resources in engineering projects

Communication Skills

- Coordinated math and science tutoring sessions with high school students, guided by consistent and effective communication skills, the student's performance had improved over the course of the tutoring sessions.
- Experience as a team member in a patent project for redesigning an all-season tire to an all-road tire; collaboration, brainstorming and individual work, were key elements in creating a sufficient patent
- Presented technical information on tire technology while working with a team of students in electrical and mechanical engineering streams, imparting technical knowledge in a manageable format to a non-technical audience
- The Royal Canadian Legion Remembrance Contest Winner, I wrote a poem as an assignment for Remembrance Day in Canada, this is a day commemorating the lives that were sacrificed in military service fought in Canada, my literary skills and creativity were celebrated and won me an award and recognition for my work

Technical Skills

- Developed a program for the Game of Life simulation, applying mathematical algorithms for problem solving and analysis in MATLAB
- Conducted electrical analysis (harmonic distortion levels, power consumption, etc.) on components such as LEDs, BJTs, and capacitors using an oscilloscope and voltmeter

• Developed an alarm clock in Verilog HDL modeling language, in a group project, assigned specific tasks to two individuals, and wrote a final report on the design of the code

• Proficient with years of experience in data structures and algorithms using C, C++, Python, & Java

• Exceptional experience in graphics manipulation using the OpenGL library in C++

• Extensive knowledge and experience using HTML and Javascript

• Written in a variety of IDEs such as BlueJ, Visual Studio, Eclipse, Notepad++, Arduino, Oracle Virtual Machine (Linux and Windows), Vivado

• Experience using Linux while compiling algorithms in C++ and Java while running a Virtual Machine

• Adobe Photoshop & Illustrator experience for personal creative endeavors, for refining design skills for future use in the business field to expand potential opportunities

APPLIED PROJECTS

BLOG WEBSITE (2025)

Purpose

It's an integrated content management system in which the users can create, publish, and engage with the blog posts. The primary aim is to create an online collaborative community that enables knowledge sharing along with the user interaction through comment and feedback mechanisms.

Scalability and Infrastructure

- Framework

- *Django:* Selected for its robust architecture and scalability. The platform is engineered to accommodate feature expansion (such as additional user roles, tagging systems, and advanced search functionalities) as both user base and content volume increase.

- Database

- *PostgreSQL:* Utilized for its efficiency in handling large datasets and concurrent user interactions, ensuring application reliability and speed at scale.

Core Technologies and Libraries

- Django & Django Rest Framework (DRF)

- Responsible for routing, data models, view logic, and template rendering. Its modularity supports maintainable and extensible codebases.

- Facilitates the development of a RESTful API, allowing integration with mobile clients and third-party applications, ensuring data accessibility beyond the core web platform.

- Bootstrap

  - Provides a responsive, cross-device front-end layout with a uniform user experience on desktop, tablet, and mobile.

- Markdown

  - Facilitates easy writing of content by enabling writers to compose posts in an abbreviated syntax that reduces barriers for non-technical writers.

- Elasticsearch

  - Enables improved search functionality, enhancing content discoverability via effective keyword, tag, and category search.

Unique Features and Differentiators

- Adaptable User Profiles

  - Users can customize their accounts, so that their experience and content choices on the website become personalized.

- Integrated Commenting System

  - Supports nested responses and offers post author and administrator controls to facilitate positive debates while maintaining content quality.

- Personalized Content Recommendations

  - Utilizing machine learning algorithms to monitor user activity and provide tailored content recommendations for greater engagement and retention.

CHAT MESSAGING PLATFORM (2025)

React | Express.js | SQL

Full-Stack Web Application | Personal Project

- Built a live chat platform with user registration, secure login, and personalized profile creation.
- Integrated private and public channel creation for each user, direct and group messaging with permanent conversation history saved in SQL databases.
- Developed responsive, interactive front-end with React, providing an identical user experience on desktop and mobile platforms.
- Implemented RESTful API with Express.js to manage authentication, users, and message routing.
- Implemented SQL relational database schema for user information, message logs, and channel information along with performance-tuned queries.
- Implemented role-based access controls and input validation to further ensure security and integrity of data.
- Using state management in React to manage live message updates efficiently.

- Structured architecture to scale for future feature enhancements like media sharing, push alerts, and search in messages.

AFRICAN BLOCKCHAIN CHAMPIONSHIP (2025)

Placed in the Top 20 Participants in the African Blockchain Championship

Overview

My project, XMR2ADA, is a blockchain integration initiative that unites Monero's privacy-focused Proof-of-Work ecosystem with Cardano's Proof-of-Stake smart contract ecosystem. The overall purpose is to provide a sustainable framework for people, especially those in underprivileged African societies, to earn and hold cryptocurrency revenue. The system provides mining, decentralized cryptocurrency exchange, and staking via a modular, extensible design—beneficial to both technological innovation and social benefit.

Core Purpose & Vision

- Accessible Crypto Income

- XMR2ADA is intended to provide the world with an open, easy procedure of participating in crypto mining and gaining. By removing hardware and technical barriers, the website strives to make earning opportunities more democratic.

- Socially Responsible Blockchain Development

- The project's structure is intended to inspire developers and organizations to harness blockchain technologies for social good, focusing on poverty alleviation and new opportunities in emerging markets.

Key Components

- XMR Mining Module

- Connects directly to Monero's mainnet and mining pools.

- Generates new Monero wallets with secure seed phrase backups.

- Supports multi-threaded mining, adjustable mining intensity, and automatic pool failover for reliability.

- Additional Insight: The mining module is optimized for low-resource environments, allowing users with basic hardware to participate.

- Automated Swap Engine

- Monitors Monero wallet balances and triggers XMR2ADA via API

- Features real-time exchange rate tracking, batch processing, and a conversion history ledger.

- Additional Insight: Automation reduces manual intervention, ensuring users can convert earnings to ADA efficiently, even with fluctuating network conditions.

- Cardano Network Module

- Generates Cardano wallets using the Cardano CLI and BIP39 mnemonic for HD wallet support.

- Compatible with mainnet and testnet, supporting multiple address types and staking pool delegation.

- Executes transactions and maintains secure blockchain connectivity.

- Additional Insight: The staking feature enables users to earn yield on converted ADA, further maximizing their crypto income potential.

Technical Architecture

- Languages & Tools

- Utilizes Python, Cardano CLI, and a suite of cryptographic libraries for cross-chain compatibility and security.

- Blockchain Networks

Monero: Proof-of-Work, privacy-preserving features (ring signatures, coin mixing).

Cardano: Proof-of-Stake, smart contract capabilities.

- Data Management

- Wallets are stored as JSON for transparency and ease of integration; blockchain data is indexed and immutable with integrity checks.

- Security

- Incorporates air-gapped key generation, hardware wallet integration, encrypted wallet storage, address whitelisting, and KYC integration points.

- Monitoring & Reporting

- Real-time mining dashboards, automated alerts, and detailed transaction and system logs provide operational visibility.

- Additional Insight

- The modular architecture ensures each component can be upgraded or expanded without affecting the overall system, supporting long-term scalability.

Scalability & Future Development

- Unified Application

    - Planned integration of all modules into a single application for seamless mining, conversion, and staking workflows.

- Server-Hosted Mainnet Sync

    - Enables lightweight client operation, making the system accessible even on low-bandwidth networks or basic hardware.

- Database Integration

    - Large-scale management of JSON-based wallets and blockchain CRUD operations for enterprise and community deployments.

- Expanded API & DeFi Capabilities

    - Future support for more swap APIs and broader DeFi integrations to expand earning and utility options.

- Additional Insight

    - These scalability features are critical for mass adoption, particularly in regions with inconsistent internet or limited computing resources.

Potential Applications

- Privacy-Preserving ADA Transactions

    - Facilitates confidential financial operations by combining Monero's privacy features with Cardano's utility.

- Automated Mining Reward Conversion

    - Streamlines the process of converting mining rewards into stakable ADA, reducing friction for users.

- Cross-Chain DeFi and Staking

    - Opens the door for users to participate in decentralized finance and staking across networks.

- Educational Tools

    - Serves as a practical resource for blockchain literacy, equipping users with hands-on financial empowerment tools.

- Additional Insight

    - By providing a full-stack solution, XMR2ADA supports both advanced users seeking automation and newcomers needing simplicity.

Impact Potential

By considerably lowering the technical and hardware limitations of cryptocurrency mining and

earning in a very major way, XMR2ADA can potentially open up new earning prospects to economically backward societies. Its business model—maintaining a good balance of technical ingenuity with a strong social responsibility spirit—can be replicated as an example to leverage blockchain for inclusive economic growth. The modular, extensible nature of the project offers a platform to grow, customize, and be adopted in mass overseas.

GAME DEVELOPMENT (February 2021 - April 2021)

3D Pong Game

- Built a unique 3D Pong game in C++ using OpenGL (Graphic Rendering API)
- Utilized Github to assist in online communication and software development between group members
- Collaborated with another group member to write a report for graded submission

INDIVIDUAL 2D GAME DESIGN (September 2020 – December 2020)

Dragon's Hollow

- Built a 2D simulated game using Object Oriented techniques, written in C++ on a Linux machine
- It runs by randomizing each iteration and outputting the states of each step until the final outcome

ARTIFICIAL INTELLIGENCE (AI) (September 2019 – April 2020)

Autonomous Vehicle

- Using reinforced & supervised learning to train an RC car, using an Arduino, to traverse a course
- Raspberry Pi & Python used to program the neural network model of the machine learning

ROBOTICS (January 2019 – September 2019)

Inverse Kinematic Articulating Robotic System

- Designing a robotic arm controlled by inverse kinematics
- Collaborating with teammates to draft PowerPoint presentations for a technical audience